

■ RAG Assistant - README

Overview

This project implements a Retrieval-Augmented Generation (RAG) based Question Answering system using LangChain, FAISS, and Streamlit. The system allows users to upload PDFs, process them into vector embeddings, and then ask questions whose answers are grounded in the uploaded documents.

■ Features

- 1 Upload any PDF file and convert it into retrievable text chunks.
- 2 Store vector embeddings locally using FAISS.
- 3 Ask natural language questions and get context-aware answers.
- 4 Built with Streamlit for an intuitive web interface.
- 5 Supports modular code organization (preprocess, embeddings, pipeline).

■ Project Structure

- 1 app.py - Main Streamlit application.
- 2 data/ - Folder containing uploaded PDFs.
- 3 src/preprocess.py - Handles PDF reading and chunking.
- 4 src/embed_store.py - Converts chunks to embeddings and stores in FAISS.
- 5 src/qa_pipeline.py - Manages retrieval and LLM-based Q&A.;
- 6 vector_index/ - Stores FAISS index and metadata files.

■■ Installation & Setup

- 1 1. Clone the repository or copy project files.
- 2 2. Create a virtual environment: ``python3 -m venv .venv``
- 3 3. Activate the environment: ``source .venv/bin/activate``
- 4 4. Install dependencies: ``pip install -r requirements.txt``
- 5 5. Run the app: ``streamlit run app.py``

■ How It Works

- 1 PDF is uploaded through the Streamlit interface.
- 2 Text is extracted and split into chunks using LangChain text splitters.
- 3 Each chunk is embedded into a vector representation using HuggingFaceEmbeddings.
- 4 FAISS stores these embeddings locally for quick retrieval.
- 5 When a question is asked, the system finds the most relevant chunks and sends them to the language model.

- 6 The model generates an answer based only on the retrieved chunks.

■ Example Questions

- 1 What is the goal of this assignment?
- 2 What are the deliverables?
- 3 Which vector database is used?
- 4 What happens if a question isn't in the document?

■■■ Developed By

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Built using Python, LangChain, HuggingFace, FAISS, and Streamlit.